

Deep learning drives next generation of crop improvement

Scientists at Aberystwyth University have developed powerful artificial intelligence (AI) tools that can automatically measure plant seeds and seed pods, creating the opportunity to accelerate the process of breeding higher-yielding, more resilient crops.

The innovation comes in a number of flavours, depending on the crop, and combines advanced computer vision and deep-learning techniques to analyse thousands of images of plant fruits. By automatically identifying and measuring features such as length, width, area and predicting volume, the system replaces the slow, manual recording of traits that has traditionally limited the speed of crop improvement. The new approach delivers what is known as high-throughput phenotyping. In one instance, MorphPod, large-scale, automated measurement of physical characteristics collected precise data from more than 300,000 individual fruits in a single study, while still attached to the plant. A related tool, DeepCanola, is designed for brassica related crops such as Rapeseed and cabbages.

Phenotyping is central to modern plant breeding, linking the visible characteristics of a plant, its phenotype, to the underlying genetics that determine how it grows, yields and resists disease. MorphPod uses image recognition to extract these measurements, allowing researchers to map variation in the morphology of pods to specific regions of the plant's DNA, helping scientists identify the

genes that influence pod size and shape. This process, known as genetic mapping, gives breeders valuable targets for selecting and combining desirable traits in future crop varieties.



“The tool represents a step-change in the accessibility of large-scale plant analysis. AI tools like the one we have developed have the potential to revolutionise how we develop new varieties of crops. By removing technical and time barriers, deep learning enables more researchers to explore plant traits at a scale that wasn't practical before.”

Kieran Atkins
PhD researcher and project lead

Deep learning refers to a type of AI model called a deep neural network, which learns complex patterns in data from large numbers of training examples. In this case, the network is trained to recognise the visual features of seed pods in photographs and to calculate

their dimensions automatically. The result is consistent, unbiased data that can be used for detailed statistical analysis.

Professor John Doonan at Aberystwyth University, said: “The breakthrough demonstrates how advanced imaging, and AI can transform crop science. Deep learning can now provide data with the quality and accuracy required for genetic analysis and breeding. This is an important step toward scalable, data-rich phenotyping that supports more predictive approaches to crop improvement.”

Although the technology was first tested on a small model plant commonly used in laboratories, the same methods have proven effective for agricultural species such as oilseed rape, cabbages, oats, barley and wheat. The team has made the MorphPod and DeepCanola software freely available online so that other research groups around the world can adapt it for their own crops.

This fusion of technology and biology is helping to future-proof farming, with Welsh research playing a key role in shaping the next generation of sustainable agriculture.



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